Question Bank-726

B.Sc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING

FIRST YEAR SECOND SEMESTER EXAMINATION, 2019

[According to the New Syllabus]

CSE-510223

(Discrete Mathematics)

Time 3 hours

Full marks 80

N.B. The figures in the right margin indicate full marks. Answer any four questions.

1. (a) What is proposition? Find the negation of the proposition "Today is Friday" and express this in simple English.
(b) What is tautology? Show that $(p \wedge q) \rightarrow (p \quad v)$ is a tautology.
(c) Express the statement "Every student in this class has studies calculus" as a universal quantification.
(d) What are the truth values of the propositions R (1, 2, 3) and R (0, 0, 1)?

2. (a) Define power set. What is the power set of the set {0, 1, 2}?
(b) Using set builder notation and logical equivalences show that $\overline{A \cap B} = \bar{A} \cup \bar{B}$.
(c) Define one-to-one and onto functions with examples.
(d) Translate the following statements into logical expressions: "Some students in this class has visited Mexico" and "Every student in this class has visited either Canada or Mexico" using quantifiers.

3. (a) Define rules of inference. Write down the basic rules of inference.
(b) Using mathematical induction show that, $1 - 2 + 2^2 + \dots\dots + 2^n = 2^{n+1} - 1$ for all nonnegative integers n.
(c) Define the Sum Rule and the Product Rule. How many different bit strings are there of length seven?
(d) Simplify the sum-of-products expansions using Karnaugh maps :
(i) $xy\bar{z} + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}\bar{z}$.
(ii) $x\bar{y}z + x\bar{y}\bar{z} + \bar{x}yz + \bar{x}\bar{y}z + \bar{x}\bar{y}\bar{z}$.

4. (a) Write a recursive procedure for Ackermann function, use the definition of the Ackermann function to find A (1,3).
(b) Find the matrix representation of the relations SoR where the matrices representing R and S are $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and

$$M_S = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

(c) How can the final exams at a University be scheduled so that no student has two exams at the same time. [For instance, suppose the courses are numbered 1 to 7 and following pairs of cources have common students : (1, 2), (1, 3), (1, 4), (1, 2), (2, 3). (2, 4), (2, 5), (2, 7), (3, 4), (3,6), (3, 7), (4, 5), (4, 6), (5, 6), (5, 7) and (6, 7)).
(d) If G is a connected planar simple graph with e edges and 5 v vertices, where v >= 3 then e <= 3v - 6.

5. (a) Define multigraph and pseudograph with examples.
(b) Construct BST of the following values 14,3, 4, 12, 14, 15, 2, 8, 2, 7, 9, 16, 6, 20.
(c) Draw the Hasse diagram representing the partial ordering 5 {(a, b) a divides b) on {1, 2, 3,4, 6, 8,12}.
(d) Evaluate the prefix expression a te + -*235^234.

6. (a) State Euler's theorem and prove it using required diagram.
(b) Describe the universality of NAND & NOR gate.
(c) What is the chromatic number of this tow graph-

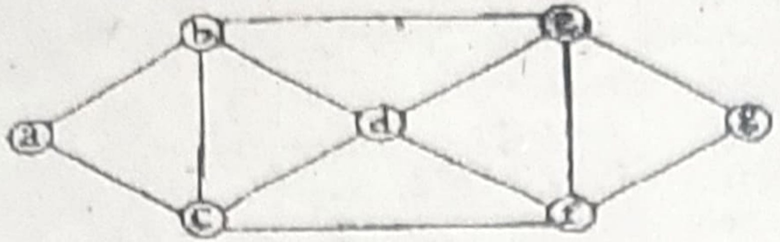

Fig. G

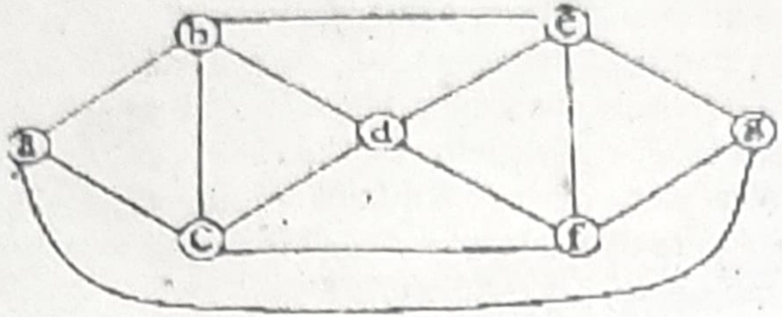

Fig. H

(d) Determine the order in which a pre-order traversal visits the vertices-